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Revealing Stereotypes: Evidence from Immigrants in Schools†

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We study how people change their behavior after being made aware of bias. Teachers in Italian schools give lower grades to immigrant students relative to natives of comparable ability. In two experiments, we reveal to teachers their own stereotypes, measured by an Implicit Association Test (IAT). In the first, we find that learning one's IAT before assigning grades reduces the native-immigrant grade gap. In the second, IAT disclosure and generic debiasing have similar average effects, but there is heterogeneity: teachers with stronger negative stereotypes do not respond to generic debiasing but change their behavior when informed about their own IAT. (*JEL* D91, I24, J15, J45)

Economists have studied discrimination toward minority groups since at least Becker (1957) and have more recently discussed how biased judgment toward specific individuals may be induced by stereotypes (Bordalo et al. 2016; Bertrand and Duflo 2017).[1] Stereotypes can be thought of as overgeneralized representations of characteristics of certain groups that allow for an easy and efficient processing of information, but they may also lead to self-fulfilling prophecies by influencing the behavior of stigmatized groups. Individuals exposed to negative stereotyping

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[1] A non-exhaustive list of papers addresses discrimination by employers (Bertrand and Mullainathan 2004), police officers (Fryer 2019; Coviello and Persico 2015; Knowles, Persico, and Todd 2001), referees (Price and Wolfers 2010), courts (Dobbie, Goldin, and Yang 2018; Alesina and La Ferrara 2014), and teachers (Figlio 2005; Botelho, Madeira, and Rangel 2015). For a review of theoretical and experimental results, see Altonji and Blank (1999) and Bertrand and Duflo (2017).

toward their own group may experience reduced effort, self-confidence, and productivity (Glover, Pallais, and Pariente 2017; Carlana 2019). Several organizations, including universities, corporations, and police departments (especially in the United States and Canada), are currently promoting interventions to mitigate discriminatory behavior by increasing employees' awareness of their implicit stereotypes.[2] However, there is limited causal evidence on the success of these policies (Bohnet 2016; Lai, Hoffman, and Nosek 2013).

We study this problem in a context where the detrimental effects of stereotyping are particularly serious: children exposed to teachers' stereotypes may be discouraged from investing in human capital (Rosenthal and Jacobson 1968; Papageorge, Gershenson, and Kang 2020; Carlana 2019). Immigrant students in Italian schools receive lower grades from their teachers compared to native students with the same performance in standardized tests, and we relate this gap in grades to teachers' implicit stereotypes. We experimentally evaluate the effects of revealing to teachers their own stereotypes and find that this leads to a change in their grading behavior, reducing the immigrant-native gap compared to a group of teachers who do not receive any information (pure control). In a second experiment, we delve deeper into the mechanisms behind the change to understand the importance of learning about *one's own* bias relative to learning about bias in general. We do not find differences on average, but the effects are heterogeneous: teachers with stronger negative stereotypes adjust their behavior when they are informed about their own bias but do not react to a generic debiasing message informing them of the presence of bias toward immigrants in society and in schools.

The case of Italy is interesting for at least two reasons. First, mass immigration is a relatively recent phenomenon, and Italy has experienced one of the highest increases in the share of immigrants over the past few years, which has fueled anti-immigrant sentiments (Alesina, Miano, and Stantcheva 2023). Second, in the Italian education system, middle school is a critical juncture, at the end of which students get tracked into different types of high schools, which affects their future education and work prospects (Carlana, La Ferrara, and Pinotti 2022a). This type of tracking is similar to that of most other European countries. The attitudes of middle school teachers could thus have important long-term effects on students' educational and professional careers.

We use two unique datasets. The first combines administrative data on students with original survey data from a sample of over 1,300 teachers in northern Italy, collected in person during a field experiment. The second dataset includes survey data from a sample of around 200 middle school teachers, collected online and embedded into a lab-in-the-field experiment where teachers were required to evaluate students' tests.

In both datasets, we measure teachers' stereotypes using the Implicit Association Test (IAT). This is a computer-based tool developed by social psychologists, designed to minimize the risk of social desirability bias in self-reported answers (Greenwald and Banaji 1995). Previous research in social psychology has highlighted a number of limitations, including weak predictive power and potential manipulation (Blanton

[2] For example, employees are advised to take an Implicit Association Test to increase awareness about one's implicit associations on race and gender. Among others, Harvard University strongly encourages "every search committee member to take at least one Implicit Association Test (IAT)" (https://faculty.harvard.edu/recruitment-best-practices), and Starbucks has promoted a "racial bias training" for all employees (https://stories.starbucks.com/press/2018/starbucks-to-close-stores-nationwide-for-racial-bias-education-may-29/).

et al. 2009; Oswald et al. 2013; Olson and Fazio 2004; Fiedler and Bluemke 2005; Cvencek et al. 2010). Despite that, it is increasingly used by social scientists to measure stereotypes both in the lab and in the field (Rooth 2010; Bertrand and Duflo 2017; Corno, La Ferrara, and Burns 2022; Glover, Pallais, and Pariente 2017; Reuben, Sapienza, and Zingales 2014). Our IAT data show that teachers generally hold strong, negative stereotypes toward immigrant students. According to the metrics proposed by Greenwald et al. (2009), around 70 percent of teachers in our field experiment and 80 percent in our online experiment exhibit "moderate to severe" stereotypes, and almost all of them exhibit some degree of negative stereotypes toward immigrants.

As a first step in the analysis, we establish that, holding constant the performance on standardized blindly graded tests, immigrant students receive lower grades than natives when they are graded by their teachers in a nonblind way. Furthermore, we correlate bias in grading with teachers' IAT and find that higher IAT scores—indicating more negative stereotypes toward immigrants—are associated with lower grades to immigrant students at the high end of the distribution, i.e., for high-performing immigrants. Teachers' IATs are uncorrelated with the grades given to native students.

We then move to the main contribution of the paper, that is, the impact of revealing stereotypes. We conducted two experiments. The first is a field experiment in around 100 schools, where we randomized the timing of the feedback on own IAT. In half of the schools (randomly selected), teachers were informed of their IAT score shortly before end-of-semester grading, while in the remaining half they were informed shortly after. We find that teachers who received their IAT score "early" gave higher grades to immigrant students relative to native ones.

Our second experiment was run online with a different group of Italian middle school teachers and complements the first one in two ways. First, it allows us to test whether learning about one's own bias has any additional effect relative to learning about generic bias in society; second, we gain a better understanding of individuals' updating process regarding their own bias. As in the field experiment, teachers in the online sample took an IAT measuring stereotypes against immigrants. Immediately after they took the test, we asked them to predict their own stereotypes and then provided feedback on IAT scores to only half of the teachers, randomly selected. Different from our field experiment, all teachers received a generic debiasing message. A few weeks after this initial session, we asked teachers to grade ten tests, randomly ascribed to native-sounding or immigrant-sounding student names. The grade given to these tests is used as the main outcome for the online experiment. We find that, on average, the personalized feedback does not increase grades assigned to immigrants versus natives, relative to generic debiasing. However, there are significant heterogeneous treatment effects: teachers with stronger implicit bias do not react to generic debiasing, but they decrease their gap in grading when provided information on their own IAT. The effect is driven by teachers who did not expect the feedback they received, hence, those who update based on new information. This is consistent with the possibility that at least part of the bias in grading was due to people being unaware of their implicit bias.

Overall, the results of our experiments suggest two things. First, from the field experiment, we learn that providing individualized feedback works on average compared to the status quo (no feedback). This is important for policy, and it could be applied at scale. Second, our online experiment suggests that IAT feedback does not work better than a generic debiasing message for teachers with mildly negative stereotypes, but it works significantly better for teachers with strongly negative stereotypes. Depending on whether the policymaker's objective is to correct the strongest biases or to work on other parts of the distribution, the IAT feedback may or may not be preferred to generic debiasing.

Our work is related to several strands of literature. First, we contribute to the recent economics literature emphasizing the importance of considering implicit bias when analyzing discriminatory behavior (Avitzour et al. 2020; Bursztyn et al. 2024; Van den Bergh et al. 2010; Guryan and Charles 2013; Corno, La Ferrara, and Burns 2022; Bertrand and Duflo 2017). Glover, Pallais, and Pariente (2017) provide evidence that exposure to managers with stronger implicit bias negatively affects the work performance of minorities. Reuben, Sapienza, and Zingales (2014) show in a lab experiment that the gender-science IAT predicts employers' biased expectations against women, while Carlana (2019) shows that teachers' stereotypes affect the gender gap in math, track choice, and self-confidence in math for girls in middle school. Research in social psychology and medicine has examined individuals' emotional responses when provided feedback about their own implicit associations, showing that people tend to react defensively—for instance, by questioning the validity of the IAT (O'Brien et al. 2010; Howell, Gaither, and Ratliff 2015; Sukhera et al. 2018). However, none of these papers investigates whether revealing people's own stereotypes to themselves has an impact on discriminatory behavior toward others. In this respect, our paper also differs from Pope, Price, and Wolfers (2018), who show that racial bias among professional basketball referees disappears after the media calls attention to the results of an academic study highlighting bias. In their study, referees learn about existing bias in the profession, not about their own.

We also contribute to the literature on teacher bias, which finds that teachers' expectations are often biased against minority students. This behavior may lead to a self-fulfilling prophecy, with students internalizing negative expectations and ultimately behaving in the direction predicted by the biased beliefs (Papageorge, Gershenson, and Kang 2020; Jussim and Harber 2005; Rosenthal and Jacobson 1968). A few previous studies compare teacher-assigned (nonblind) grades and standardized (blind) test scores across minority and nonminority students (Botelho, Madeira, and Rangel 2015; Burgess and Greaves 2013; Hanna and Linden 2012; Van Ewijk 2011) and across genders (Lavy and Sand 2018; Lavy 2008; Lavy and Megalokonomou 2024; Terrier 2020). We add to this literature by using the IAT as a direct measure of teachers' stereotypes, which allows us to trace a stronger link between grading gaps and bias even in the presence of unobserved characteristics that may lead students to perform differentially in blindly versus nonblindly graded tests. Furthermore, none of the above papers test the effectiveness of remedial interventions to mitigate bias in the schooling context.

Finally, our results speak to a recent literature on how to reduce bias. Lai et al. (2014) underline that interventions providing counterstereotypical exemplars and strategies to override biases are the most effective in reducing implicit racial prejudice. The impact of diversity training on behavior change is discussed by Chang et al. (2019), while recent awareness-raising campaigns on gender bias have been studied by Boring and Philippe (2021); Mengel (2021); Carnes et al. (2015); and Devine et al. (2017). The interventions on bias mitigation toward immigrants have mainly focused on providing information about immigrants and have estimated the effect on attitudes (Grigorieff, Roth, and Ubfal 2020; Hopkins, Sides, and Citrin 2019) and support for immigration policies (Facchini, Margalit, and Nakata 2022; Alesina, Miano, and Stantcheva 2023). An additional group of interventions focuses on promoting intergroup contact (Allport 1954, p.276; Lowe 2021; Corno, La Ferrara, and Burns 2022). As suggested in a recent meta-analysis by Paluck, Green, and Green (2018), "the absence of studies addressing adults' racial or ethnic prejudices [is] an important limitation for both theory and policy."

The remainder of the paper is organized as follows. In Section I we provide background information on the grading system in Italian middle schools. Section II describes our data and the experimental design, and Section III contains descriptive evidence on implicit stereotypes and grading. Section IV presents our results, and Section V concludes.

I. Institutional Background

A. *The Italian Schooling System*

Education in Italy is free for all children and is compulsory from the ages of 6 to 16. The schooling system is organized as five years of primary school, three years of middle school, and five years of high school. Students are assigned to the same class for all subjects, and they interact with the same set of peers within each type of school. In middle school, which comprises grades 6 to 8, students are usually taught by the same teachers for all three years, and they spend at least six hours per week with the math teacher and five hours with the literature and grammar teacher. Teachers are assigned to schools by the Italian Ministry of Education, and their allocation is determined by seniority: teachers with more experience can teach at schools that are higher in their preference ranking and tend to work close to their hometown and away from disadvantaged areas (Barbieri, Rossetti, and Sestito 2011). Students are assessed continuously with written and oral exams in each subject, and they receive end-of-semester grades in January and June. These "final" grades are discrete variables ranging between 3 and 10, with 6 being the pass grade. Thus, end-of-semester grades may incorporate significant discretion of the teachers.

In addition to teacher evaluations, standardized tests in math and reading are administered by the National Institute for the Evaluation of the Education and Training System (INVALSI) to all Italian students at the end of middle school (grade 8). INVALSI tests mainly consist of multiple-choice questions or short answers, which are blindly graded following a precise evaluation grid.

At the end of middle school, students must choose between three high school tracks: academic oriented (liceo), technical, and vocational. Academic and technical schools offer significantly better educational and employment prospects than vocational schools (Carlana, La Ferrara, and Pinotti 2022a).

B. *Immigrants in Italian Schools*

In the last two decades, the share of immigrant children (i.e., children without an Italian citizenship) in Italian schools has increased from less than 1 percent in 1998 to 10 percent in 2018, with a higher concentration in the northern part of the country and big cities. Immigrant students come from diverse geographic backgrounds, with the most represented nationalities being Romanian, Albanian, Moroccan, Chinese, Filipino, and Indian (see online Appendix Table A.1). Currently, about 65 percent of immigrant children are born in Italy, but they can obtain Italian citizenship only after turning 18 and are subject to rather stringent conditions.[3] Throughout the paper, immigrant students are defined according to their citizenship: they include first-generation students born abroad and second-generation students born in Italy from parents who are not Italian citizens.

Immigrant students have, on average, lower performance than native students in Italian schools (Carlana, La Ferrara, and Pinotti 2022a), and the same is true in most other destination countries (OECD 2014). Of course, this may at least in part reflect language barriers and parental investment given that, on average, they typically come from a lower socioeconomic background, but it may also partly reflect discrimination by teachers.

II. Data and Experimental Design

A. *The IAT*

We measure implicit stereotypes toward immigrants using the Implicit Association Test. The test requires categorizing words to the left or to the right of a computer screen, and it measures the strength of the association between two concepts based on response times. The underlying idea, as conceived by Donders (1969) and Greenwald, McGhee, and Schwartz (1998), is that the easier the mental task, the faster the response production.

The version of the IAT that we developed for our study requires associating immigrant and native names (e.g., Fatima and Francesca) with positive and negative adjectives in the schooling context (e.g., smart and lazy). Labels and categories are in the top corners of the screen, names and adjectives randomly appear at the center of the screen, and subjects are asked to categorize the words as quickly as possible. If respondents hold negative stereotypes against immigrants, they should react more slowly when the label "immigrant" is associated with positive adjectives compared to when it is associated with negative ones because positive associations are less natural to them. The IAT measures stereotypes using the difference in reaction times between rounds in which native-sounding names and negative adjectives appear on the same side of the screen and rounds in which immigrant-sounding names and negative adjectives appear on the same side.

Starting from the continuous IAT score produced by the test, one can define a categorical measure based on conventional thresholds recommended by Greenwald et al. (2009). In particular, the negative association with immigrant names is absent when the IAT score is positive but below 0.15, "slight" when it is between 0.15 and

[3] Like most other European countries—and unlike the United States—Italy follows the principle of ius sanguinis; i.e., citizenship is determined by the nationality of one's parents. There is a limited time window (1 year) to apply for Italian citizenship after turning 18, and the candidate citizen must be able to prove continuous residence in Italy during the previous years.

0.35, and "moderate to severe" when it is above 0.35. Negative values of these same thresholds define the strength of positive associations.

In the field experiment, each teacher in our survey completed two immigrant-native IATs, one using male names and one using female names, and the order of the IAT with male and female names was randomized at the individual level. In the online experiment, we administered the IAT using a mix of male and female names of immigrant and native students. This allowed us to minimize the duration of the baseline survey (a key aspect given the online setting) and to calculate only one IAT score per teacher. Further details on the IATs that we administered are available in online Appendix B.

While the IAT is widely used (Green et al. 2007; Arcuri et al. 2008; Nosek et al. 2009; Monteith, Voils, and Ashburn-Nardo 2001), previous research in social psychology has highlighted a number of limitations (Olson and Fazio 2004). First, some argue that the IAT has a weak predictive power (Blanton et al. 2009; Oswald et al. 2013; Meissner et al. 2019) and, in particular, that it does not predict behavior better than explicit measures (Axt, Bar-Anan, and Vianello 2020; Schimmack 2021). However, most of these studies refer to experiments with a limited number of subjects and do not have information outside the lab on whether individuals with stronger implicit associations are actually biased in their interactions. Recent papers in economics have shown correlations of IAT scores with real-world behavior, including call-back rates of job applicants (Rooth 2010), job performance of minorities (Glover, Pallais, and Pariente 2017), and teachers' track recommendations (Carlana, La Ferrara, and Pinotti 2022b).

The second main concern with the IAT is that subjects may fake the test by voluntarily slowing down or speeding up on specific blocks or strategically increasing errors (Fiedler and Bluemke 2005; Cvencek et al. 2010). However, this type of manipulation would require a deep knowledge of the test, which is unlikely within our sample of teachers, as the IAT is not widely known in Italy. Furthermore, the improved scoring algorithm that we use (Greenwald, Nosek, and Banaji 2003) discards observations characterized by abnormal reaction times.

Third, some researchers argue that the IAT measures social constructs such as salience of attributes (Rothermund and Wentura 2004), familiarity with the concepts it quantifies, and, more generally, cultural stereotypes rather than "personal animus" (Arkes and Tetlock 2004; Karpinski and Hilton 2001; Mitchell and Tetlock 2017; Tetlock and Mitchell 2009). However, these possibilities have been addressed empirically (Nosek and Hansen 2008; Olson and Fazio 2004; Ottaway, Hayden, and Oakes 2001; Rudman et al. 1999; Dasgupta and Greenwald 2001), and past research in social psychology suggests there is no reason why familiarity and attitudinal evaluation should be unrelated since familiarity breeds liking (Jost 2019).

A fourth concern is that the IAT may capture unstable characteristics that vary over time (Dasgupta and Greenwald 2001; Bar-Anan and Nosek 2014; Gawronski et al. 2017). However, social psychology theory establishes that attitudes are intrinsically dynamic (Banaji 2004; Hardin and Banaji 2013). Moreover, the IAT exhibits a higher (within-person) test-retest reliability than other response-latency measures commonly used in psychological research, including Stroop and priming tasks (Bar-Anan and Nosek 2014; Jost 2019).

Overall, we acknowledge that the IAT may be a noisy measure of stereotypes, but it has the advantage of (i) avoiding social desirability bias present in explicit responses on socially sensitive topics (Greenwald et al. 2009) and (ii) capturing implicit associations that may be unknown to the individual but may nevertheless affect their interaction with stigmatized groups (Bertrand, Chugh, and Mullainathan 2005).

B. *The Field Experiment*

In our first experiment, we administered an IAT to a large sample of teachers of grade 8 students and revealed to half of them their own IAT score just before end-of-semester grading, while the other half received the same information shortly after end-of-semester grading. We then compared the grades given to immigrants and natives between the two groups of teachers.

The experiment took place in five large cities of northern Italy—Milan, Brescia, Padua, Genoa, and Turin—during the first part of the 2016–2017 school year. In September 2016, all middle schools in these cities enrolling at least 20 immigrant students in grade 6 (as of 2012) were invited to participate in a survey titled "The role of teachers in high school track choice." We intentionally avoided mentioning immigrants and immigration-related issues to prevent sample selection on attitudes toward immigration. Out of 145 schools invited, 102 accepted to take part in the project.[4,5]

The survey was addressed to all math and literature teachers in grade 8, and it consisted of two parts. In the first part, teachers completed two immigrant-native IATs, one with male names and one with female names, as described in the previous section. In what follows, we use the average of the two.[6] The second part of the questionnaire elicited information on respondents' socioeconomic characteristics, teaching experience, explicit bias toward immigrants, and criteria followed to advise students on high school track choice.[7]

On average, 80 percent of the teachers in our 102 schools completed the survey, yielding a sample of 1,384 teachers. This is the main sample used for estimating

[4] It is useful to discuss if and how these 102 schools differ in terms of student and teacher characteristics. We cannot provide balance tables of the characteristics of students in the 102 schools compared to the 43 schools that did not participate, as we do not have the code to identify those 43 schools from the pseudo-anonymized dataset of Italian schools. However, in online Appendix Table A.2, we compare students in our experimental sample with all students in Italian schools (column 1) and all other students in the selected provinces (column 2). Schools in our sample are comparable in terms of gender composition but have a higher share of immigrants than other schools, as should be expected given the selection criteria for our study. This also implies some differences in socioeconomic characteristics correlated with immigrant status; however, the standardized differences are very small for all variables.

[5] Data and documentation for both experiments are available on the data depository Alesina et al. (2024).

[6] The correlation between the two continuous IAT scores is 0.28. However, based on the categories we communicated to teachers, 76 percent of them received a consistent message (either biased in both or unbiased in both): despite the noise in the measurement, the test is accurately capturing individuals' implicit associations between immigrants/natives and positive/negative adjectives. Teachers also completed a gender-science IAT (see Carlana 2019) that we do not use in this paper. The order of the IATs was randomized across individuals, but the two immigrant-native IATs were always presented one after the other. The correlation between each immigrant-native IAT and the gender-science IAT is lower (0.06) compared to the correlation between the two immigrant-native IATs (0.28), suggesting that the IAT is not merely capturing the ability to complete the test.

[7] The questionnaire was administered during meetings held in school buildings. Our enumerators gave each teacher one tablet to complete the survey autonomously but remained available in the room to answer questions or help with tablets if requested. Teachers who agreed to take part in the survey gave written informed consent. The time to complete the survey was around 30 minutes, and participants did not receive any compensation.

the relationship between teachers' IAT and grading of immigrant students. To this purpose, we obtained both teacher-assigned grades and standardized test scores in grade 8 for all students taught by these teachers between school years 2011–2012 and 2016–2017.

Within the 102 schools, 65 schools comprising 533 teachers in grade 8 were surveyed before the end of the first semester (i.e., end of January) due to logistical reasons and are therefore included in the experimental sample, while others were interviewed after the end of January.[8] This is the experimental sample used for estimating the effect of revealing IAT scores on grading behavior. We offered to all teachers in this subsample of schools the possibility of receiving feedback on their IAT score, and more than 80 percent of teachers chose to receive it. Online Appendix Table A.5 shows that there is no significant correlation between the decision to receive the feedback and several teacher characteristics, including implicit or explicit biases against immigrants.[9]

Feedback was provided over email. Teachers received a brief description of the IAT and were told whether their association between immigrant names and good/bad adjectives was "slight," "moderate," or "strong" based on the thresholds identified by Greenwald et al. (2009) and discussed in Section IIA. Each teacher received their score from two IATs: one using male names of natives and immigrants and one using female names. Teachers were assured that these results would not be shared with anyone. The detailed text of the email is reported in online Appendix B.B2.

We randomized the timing of the feedback across schools. In half of the schools ("treatment"), teachers received the feedback before end-of-semester grading, i.e., by the end of January 2017. In the remaining schools ("control") teachers received the feedback within two weeks after end-of-semester grading. This implies that all teachers (in both the treated and control groups) learned about their IAT by mid-February, which prevents us from studying the long-term impact of our intervention. We chose to randomize at the school level, rather than at the teacher level, to avoid contamination.

Leveraging the randomization, we can estimate the effect of revealing IAT on grading behavior by comparing the grades assigned by teachers in the treated and control groups to immigrant and native students. The grades given by teachers at the end of the first semester are normally the arithmetic mean of previously assigned scores in written and oral exams, where teachers have substantial power to decide whether to round the score up or down. We expect that our intervention may affect this discretionary choice of the teacher. We used grades available from administrative registries so that teachers were unaware that we could observe their grades, thus reducing the risk of experimenter demand effects.

Figure 1 illustrates the timeline of the survey and experiment, as well as the periods covered by the data on standardized test scores and teacher-assigned grades. Note that

[8] Online Appendix Table A.3 compares the two sets of teachers, while online Appendix Table A.4 compares the characteristics of their students. In both cases, the two groups are comparable.

[9] Instead, there is a significant correlation with how much "in a hurry" the respondent was. A survey completion time of more than 20 minutes (33 percent above the mean) is associated with a 5 percentage point increase in the probability of consenting to receive the feedback. Similarly, those who completed only the IAT and not the rest of the survey were almost 10 percentage points less likely to request feedback. These correlations do not survive when including school fixed effects, which explains a substantial share of the variation in the choice of receiving feedback, as shown by the R^2 in online Appendix Table A.5.

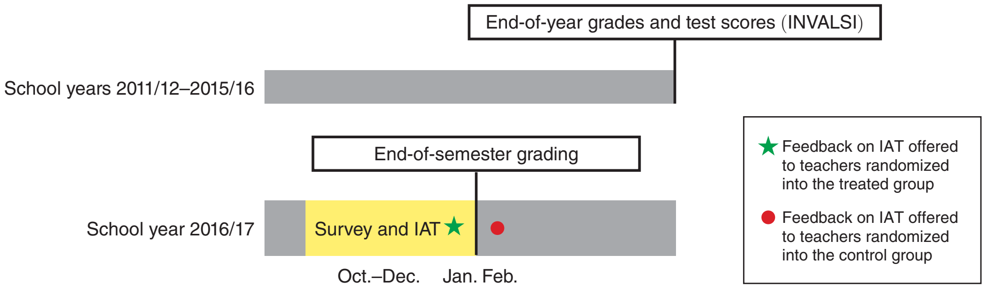

Figure 1. Timeline of Data Collection for the Field Experiment

Notes: This figure shows the timeline of the data collection, survey, and field experiment. As described at length in Section II, we obtained administrative data on end-of-year teacher-assigned grades as well as on standardized, blindly graded test scores for school years 2012–2013 through 2015–2016. During the first semester of the 2016–2017 school year (October–January), we administered the survey and the IAT to all teachers in our sample. In January 2017, before end-of-semester grading, we sent feedback about teachers' own IAT scores to a random group of teachers. All other teachers were allowed to see their score after the end-of-semester grading (i.e., February 2018).

when we study the role of teacher stereotypes in grading, we use *end-of-year* grades (i.e., in June), as these are contemporaneous to the (blindly graded) INVALSI test scores in grade 8, which are essential for this type of analysis.[10] In contrast, when we estimate the effect of revealing IAT scores, we use *end-of-semester* grades (i.e., in January), for ethical reasons: these grades are not decisive for students' careers, and hence, we minimize the possibility of our intervention harming students' outcomes. Unfortunately, INVALSI tests are *not* administered midyear, which implies that in analyzing the impact of our experiment, we cannot control for the INVALSI score. This, however, does not affect our ability to estimate the effects of the intervention, given randomization.

C. *The Online Experiment*

Our second experiment aims to isolate the effect of the *unexpected* component of bias revelation and to compare the effects of revealing one's own bias to those of a more generic debiasing message. From December to January 2021, we invited 595 teachers to an online survey, which was completed by 179 teachers from 74 different schools.[11] This baseline survey included the immigrant-native IAT described in Section IIA, together with a short questionnaire collecting basic demographic characteristics. After having completed the IAT, participants were asked whether they expected to have no bias against immigrants or a "slight," "moderate," or "strong" bias. We classify respondents as *underestimating* their own bias

[10] Note that knowledge of our study could not affect the behavior of teachers toward the cohorts of children used for this part of the analysis given that they graduated from middle school before our data collection.

[11] The pool of teachers we invited were part of a separate data collection for the Tutoring Online Program (TOP) described in Carlana and La Ferrara (2024). Teachers received the invitation upon completing the endline survey of TOP, with the following recruitment message: "Are you interested in completing a survey for another research project and getting a thank you voucher of 40 euros? This is for a project completely independent from TOP, aimed at understanding the way in which teachers grade assignments. Your participation in this second research will not affect the participation in the TOP program in the future. We expect that the second research project will require a total of 45 minutes, divided in two moments."

whenever this self-assessment is lower than the classification based on their actual IAT score, using the thresholds defined by Greenwald et al. (2009) and discussed in Section IIA.

After teachers completed the baseline survey, we randomized them into two groups, at the school level. The first group ("active control") comprised 88 teachers who received a generic debiasing message, with information on implicit biases in society and their potential negative impact on students. The second group ("treatment") comprised 91 teachers who received the generic debiasing message plus information on their own IAT score.[12] The detailed content of the two messages is reported in online Appendix B.B2.

Teachers received the debiasing message and, if applicable, their IAT score by email at the end of January 2021. Approximately three weeks later, we contacted them again and asked them to grade ten short tests in their subject (alternatively, math, literature, or English). We randomized across teachers the name of the student reported in each answer, between typical immigrant names (two tests with female names, two with male ones) and typical native names (three tests with female names, three with male ones). The tests were prepared by consultants hired by our team, who were teachers in middle schools outside our sample. The same consultants provided sample answers of varying quality, corresponding to different test grades. These answers are the ones that were submitted for grading to the teachers in our online experiment.[13] Online Appendix Figure A.1 shows that there is a very high correlation between the intended grade according to the consultants who prepared the tests and the average grade assigned by the teachers who participated in our experiment.

D. *Descriptive Statistics*

IAT Score and Teacher Characteristics.—Figure 2 plots the distribution of the IAT score across teachers in the field and in the online experiments (panels A and B, respectively). The vast majority of teachers have negative stereotypes toward immigrants (i.e., an IAT score greater than 0.15), with no relevant differences between literature and math teachers. About 80 percent of teachers in the online experiment exhibit strong stereotypes (i.e., IAT score greater than 0.35), compared to 67 percent in the field experiment.

In addition, the last row of panel B of Table 1 shows that 80 percent of teachers in the online experiment underestimate their biases.[14] In general, teachers in the online experiment exhibit a higher average IAT compared to participants in the field

[12] Ideally, we would have liked to implement the experiment with three arms: Generic debiasing + IAT feedback, Generic debiasing, and Pure control. However, given the difficulties in recruiting a large enough number of teachers for the online experiment during the pandemic, we decided to prioritize the comparison between generic debiasing and IAT revelation, and we included only two treatment arms.

[13] Some examples of test questions and answers are available in online Appendix B.B3. The incomplete disclosure of the fictitious exams and names during the experiment did not have more than minimal risk for teachers. After the experiment, following the IRB protocol, teachers were informed with a debriefing message on the detailed purpose and incomplete disclosure of the experiment.

[14] In the field experiment we did not elicit teachers' priors about their IAT.

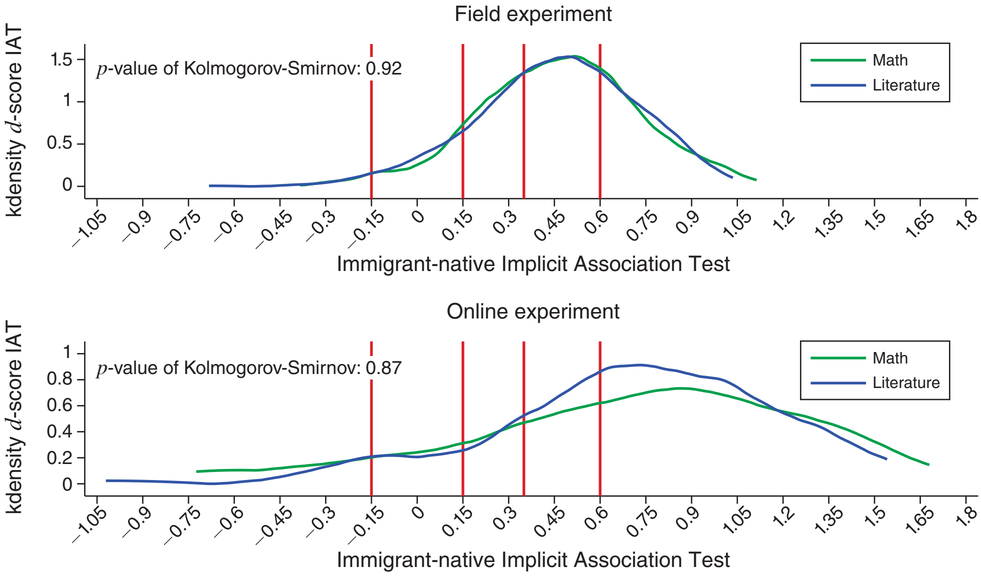

FIGURE 2. DISTRIBUTION OF THE IMMIGRANT-NATIVE IAT SCORE ACROSS TEACHERS

Notes: This graph shows the distribution of raw IAT scores for math and for literature teachers. A positive value indicates a stronger association between "natives" and "good" and "immigrants" and "bad." The first panel reports the IAT score for teachers participating in the first experiment (in person, 1,384 teachers), while the second panel shows the IAT score of teachers participating in the second experiment (online, 146 teachers). The vertical lines indicate the critical thresholds suggested by Greenwald et al. (2009) for defining different levels of bias. The negative association with immigrant names is absent when the IAT score is positive but below 0.15, "slight" when it lies between 0.15 and 0.35, and "moderate to severe" when it is above 0.35. Negative values of these same thresholds define the strength of positive associations.

experiment: 0.70 and 0.48, respectively.[15] One potential explanation for this difference is related to the different timing and implementation of the test—in person before the COVID-19 pandemic for the field experiment and remotely during the pandemic for the online experiment.

Most importantly for our purposes, the first row of panels A and B in Table 1 show that average IAT scores are balanced between the treated and control groups within each experiment, as one would expect given randomization. The remaining rows of the table show that other observable characteristics are also balanced between the two groups.[16] Not only are the differences not statistically significant, but in all cases the normalized difference (column 5) remains below the threshold of 0.25, as recommended by Imbens and Rubin (2015).[17]

[15] The mean IAT score in our experiments is slightly higher than the mean of 0.41 in the sample of Italians who decided to take the race IAT online on the website https://implicit.harvard.edu.

[16] In online Appendix Table A.7, there is only one student-level observation if both math and literature teachers of the same student participate in the experiment. For this reason, the sample includes 6,050 students, while in the analysis we will include student by teacher observations. The sample is balanced also when considering separately the students whose math teacher participates in the experiment (85 percent of the student sample) and the students whose literature teacher participates in the experiment (85 percent of the student sample).

[17] The formula for the normalized difference is $\Delta = (\bar{X}_T - \bar{X}_C)/\left(\sqrt{S_T^2 + S_C^2/2}\right)$, where $\bar{X}_T$ and $\bar{X}_C$ are the means of covariate X in the treated and control group, respectively, and S_T^2 and S_C^2 are the corresponding sample variances of X.

Table 1—Balance Table: Teacher Characteristics

	Full sample (1)	Control (2)	Treated (3)	*p*-value (4)	Norm. diff. (5)
Panel A. Teachers in the field experiment					
IAT	0.477	0.493	0.464	0.174	−0.079
	(0.261)	(0.269)	(0.253)		
Female	0.867	0.835	0.892	0.116	0.118
	(0.340)	(0.372)	(0.311)		
Teaching Math	0.494	0.506	0.485	0.397	−0.030
	(0.500)	(0.501)	(0.501)		
Advanced STEM	0.105	0.122	0.091	0.245	−0.071
	(0.307)	(0.328)	(0.288)		
Born in the North	0.665	0.679	0.653	0.577	−0.039
	(0.473)	(0.468)	(0.477)		
Age	47.455	48.114	46.929	0.406	−0.066
	(12.809)	(11.613)	(13.685)		
Full-time contract	0.826	0.802	0.845	0.313	0.080
	(0.380)	(0.400)	(0.362)		
Experience/10 years	1.955	1.967	1.946	0.881	−0.012
	(1.191)	(1.191)	(1.192)		
Children	0.702	0.696	0.707	0.836	0.017
	(0.458)	(0.461)	(0.456)		
Low edu Mother	0.448	0.468	0.431	0.434	−0.053
	(0.498)	(0.500)	(0.496)		
Middle edu Mother	0.301	0.304	0.300	0.914	−0.006
	(0.459)	(0.461)	(0.459)		
High edu Mother	0.150	0.148	0.152	0.926	0.008
	(0.357)	(0.356)	(0.359)		
Degree Laude	0.243	0.232	0.253	0.574	0.035
	(0.430)	(0.423)	(0.435)		
WVS Immigrants' Rights to Job	0.594	0.591	0.596	0.909	0.007
	(0.492)	(0.493)	(0.492)		
Observations	534	237	297		
Panel B. Teachers in the online experiment					
IAT	0.704	0.729	0.677	0.493	−0.073
	(0.502)	(0.496)	(0.510)		
Female	0.863	0.851	0.875	0.680	0.049
	(0.345)	(0.358)	(0.333)		
Born in the North	0.603	0.568	0.639	0.451	0.102
	(0.491)	(0.499)	(0.484)		
Experience	20.308	20.770	19.833	0.606	−0.062
	(10.574)	(10.389)	(10.812)		
Teaching Math	0.349	0.338	0.361	0.746	0.034
	(0.478)	(0.476)	(0.484)		
Teaching Italian	0.479	0.486	0.472	0.856	−0.020
	(0.501)	(0.503)	(0.503)		
Underestimate own IAT	0.801	0.811	0.792	0.783	−0.033
	(0.400)	(0.394)	(0.409)		
Observations	146	74	72		

Notes: The table shows the mean of the characteristics of the samples of teachers included in the final analysis (column 1), teachers in the control group (column 2), and teachers in the treatment group (column 3). Data for the field experiment are reported in Panel A and for the online experiment in Panel B. Standard deviations are in parentheses in columns 1, 2, and 3, and the *p*-value of the difference is in column 4. The last column reports the normalized difference between group averages.

Table 2—Correlation between Teacher Characteristics and IAT (Field Experiment Sample)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. IAT score (stereotypes against immigrants) in field experiment								
Female	−0.042						−0.040	−0.046
	(0.020)						(0.027)	(0.033)
Born in the North		−0.021					−0.057	−0.043
		(0.014)					(0.017)	(0.020)
Experience/10 years			−0.003				0.004	0.007
			(0.005)				(0.008)	(0.009)
WVS Immigrants' Rights to Job				−0.058			−0.045	−0.042
				(0.017)			(0.025)	(0.031)
Share of Immigrants					−0.065		−0.070	−0.212
					(0.067)		(0.067)	(0.133)
Native-Imm INVALSI(/100)						−0.040	−0.022	−0.041
						(0.083)	(0.088)	(0.119)
School fixed effects	No	No	No	No	No	No	No	Yes
Observations	1,384	1,384	1,384	1,384	779	779	779	779
R^2	0.063	0.061	0.061	0.066	0.093	0.092	0.117	0.203
Panel B. IAT score (stereotypes against immigrants) in online experiment								
Female	−0.014						−0.007	0.110
	(0.088)						(0.090)	(0.204)
Born in the North		−0.044					−0.070	−0.221
		(0.082)					(0.081)	(0.176)
Experience/10 years			0.057				0.050	0.013
			(0.036)				(0.039)	(0.072)
WVS Immigrants' Rights to Job				−0.139			−0.127	−0.190
				(0.074)			(0.078)	(0.158)
School fixed effects	No	No	No	No			No	Yes
Observations	146	146	146	146			146	146
R^2	0.006	0.008	0.020	0.023			0.037	0.444

Notes: This table reports OLS estimates, where the dependent variable is the IAT score of teachers and the unit of observation is teacher t in school s. Panel A reports the correlations for teachers in the field experiment, while panel B reports the correlations for teachers in the online experiment. We include controls for the order of IATs and for whether the blocks were presented in an order-compatible or order-incompatible way (which was randomized at the individual level). The variable "WVS Immigrants' Rights to Job" equals one for teachers believing that immigrants should have the same right to jobs as natives. "Native-Imm INVALSI(/100)" indicates the difference in average standardized test scores of native and immigrant students assigned to the teacher in the previous four years. In columns 5–8 of panel A, the number of observations decreases because information on past students is not available for all teachers; in these columns, we control for the number of observations with information available for at least three immigrant and native students.

Table 2 shows the correlation between the IAT score and other teacher characteristics, both for the field experiment (panel A) and for the online experiment (panel B). The correlation between gender, place of birth, and working experience is small and generally nonsignificant (columns 1–3). On the other hand, there is a significant correlation between IAT and explicit beliefs about immigrants, as measured by a question asking whether immigrants and natives should have equal opportunities to access available jobs (the variable "WVS Immigrants' Rights to Job" in the table, as a similar question is routinely included in the World Values Survey). Column 4 shows that respondents who agree with this statement have significantly less negative implicit stereotypes against immigrants.

In columns 5 and 6 of panel A, we test whether teachers' stereotypes reflect the relative ability of native and immigrant students to whom teachers were previously exposed. For this purpose, we collected the standardized test scores (INVALSI) of the students taught by teachers in our sample during the five years before our analysis. We could recover previous students' test scores for 779 out of 1,384 teachers, which explains the reduced sample size in columns 5–8 of panel A.[18] We find no meaningful correlation between teachers' IAT scores and the share of immigrant students they taught in the past (column 5), nor with the difference in the average test scores of past native and immigrant students (column 6). This suggests that stronger stereotypes toward immigrant students may not reflect statistical discrimination based on objective information on average group ability. The results remain qualitatively similar (with the exception of the *Northern* dummy in panel A) when we introduce all regressors at the same time and when we include school fixed effects (columns 7 and 8).

In addition, online Appendix Table A.6 shows no significant correlation between IAT score and characteristics such as having children, parents' education, and the beliefs on the reasons underlying the gap in high school track choice between native and immigrant students (e.g., ability, economic conditions, language differences, prejudice).[19]

Grades and Student Characteristics.—Figure 3 shows the distribution of teacher-assigned grades (left graph) and standardized test scores (right graph) for native and immigrant students at the end of the school year, compiled using data for all schools in our field experiment sample over the school years 2011–2012 to 2015–2016 (i.e., before our experiment). The two measures have different scales, with teacher-assigned grades ranging from 3 to 10 and INVALSI scores from 0 to 100.

The leftmost graph shows that at the end of school year, there is a substantial bunching in teacher-assigned grades at 6 ("pass"), for about 60 percent of immigrant students and 35 percent of native students. The distribution of both teacher grades and standardized test scores for native students first-order stochastically dominates that for immigrants. This gap may reflect differences in academic performance between native and immigrant students as well as other factors (e.g., gaps in diligence, behavioral issues, teacher bias in grading).[20] Online Appendix Table A.7 confirms that past grades and all other student characteristics are balanced between the treated and control group.

III. Implicit Stereotypes and Grading

In this section, we compare teacher grades between immigrant and native students, holding constant standardized test scores, and we relate differences in grading to teachers' IAT scores. Figure 4 plots the average grades assigned by teachers to

[18] We include teachers who had at least three immigrant (and native) students.

[19] The detailed questions are reported in online Appendix B.B1.

[20] Online Appendix Figure A.3 reports the distribution of teacher-assigned grades separately for math and literature. The pattern is very similar.

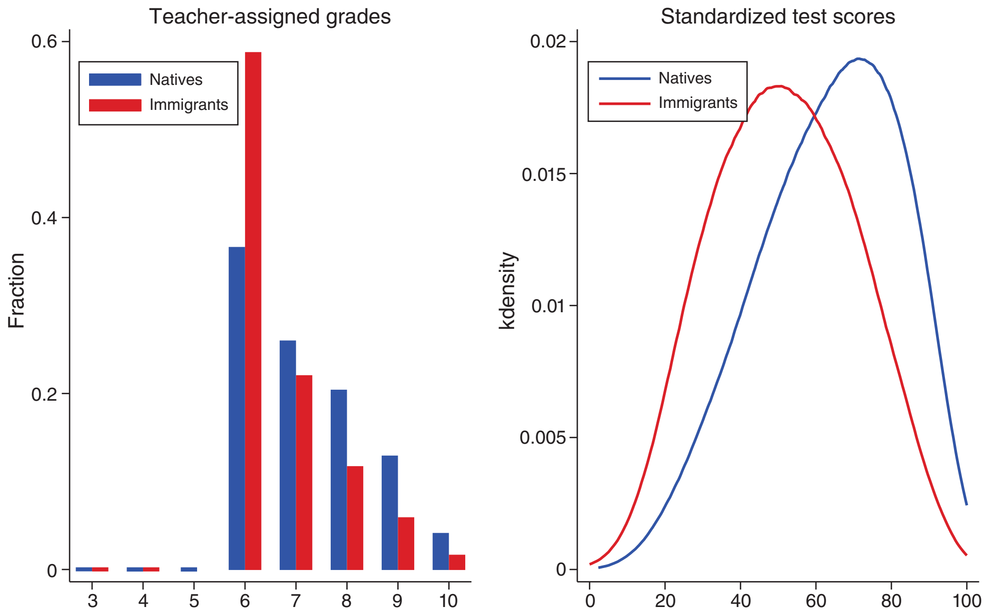

Figure 3. Distribution of Grades

Notes: The graphs show the distribution of teacher-assigned grades (left panel) and standardized test scores INVALSI (right panel). The blue bar is for native students, and the red bar is for immigrant students. For both teacher-assigned grades and standardized test scores, we report the average of math and reading scores. Students in this sample completed grade 8 between school years 2011–2012 and 2015–2016.

immigrant and native students (on the vertical axis) by quintile of the standardized test score (on the horizontal axis), with the associated 95 percent confidence intervals. Not surprisingly, students with a higher standardized test score receive, on average, a higher grade from their teacher, with a correlation of 0.56. However, conditional on obtaining the same standardized test score, immigrant students receive significantly lower grades from teachers, particularly in the upper part of the test score distribution.[21] The average gap is 0.14, comparable in magnitude to the difference explained by maternal education: controlling for the quintiles of the standardized test score, students whose mothers have less than a high school diploma receive a grade that is on average 0.21 points lower compared to children of mothers with a high school diploma or university degree.

The difference highlighted in Figure 4 may reflect teacher bias against immigrant students (see, e.g., Botelho, Madeira, and Rangel 2015; Burgess and Greaves 2013; Hanna and Linden 2012; Lavy 2008). However, there may also be other reasons why immigrant students perform better in standardized tests than in teacher-graded assignments. For instance, teachers could place greater emphasis on multidimensional competence (e.g., oral expression, behavior in class) that is not easily captured by standardized tests. To corroborate the role of teachers' implicit stereotypes, we relate differences in grading to teachers' IAT scores.

[21] Online Appendix Figure A.4 provides separate figures for math and literature. The gap is found in both subjects.

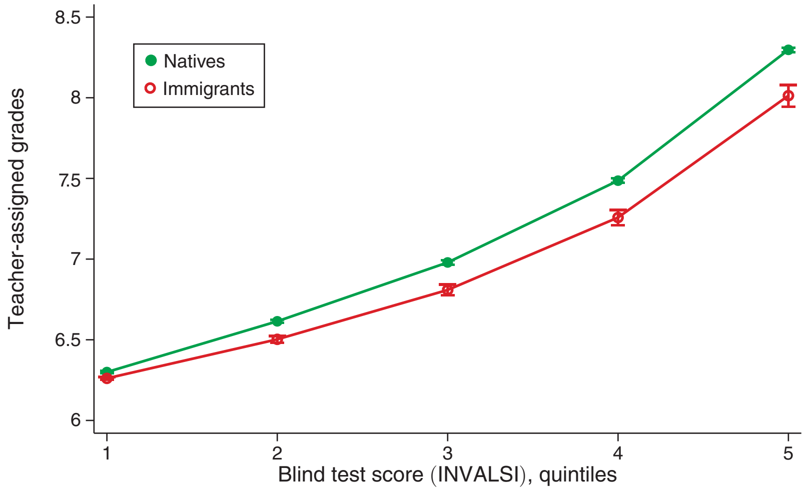

Figure 4. Teacher-Assigned Grades versus Blindly Graded, Standardized Test Scores

Notes: This graph shows teacher-assigned grades (nonblindly graded) on the vertical axis and quintiles of the standardized test score INVALSI (blindly graded) on the horizontal axis at the end of grade 8. Teacher-assigned grades are on a scale of 3 to 10, with 6 as the pass grade. The green circles and lines are for native students, while the red circles and lines are for immigrant students. Students in this sample completed grade 8 between school years 2011–2012 and 2015–2016.

Figure 5 shows the association between teachers' implicit stereotypes, as measured by their IAT score, and the grading of native and immigrant students. The black and blue solid lines represent the residuals from regressions of grades assigned to native and immigrant students, respectively, on teacher fixed effects, a cubic polynomial in the INVALSI test score, and cohort fixed effects (dashed lines represent the associated 95 percent confidence intervals). Higher values of the IAT score are associated with significantly lower grades to immigrant students, while they do not correlate with the grades assigned to native students (the black line remains flat around zero over the entire distribution of the IAT score). Table 3 quantifies the effects shown in Figures 4 and 5. Even controlling for teacher fixed effects and the cubic polynomial of the INVALSI test score, immigrant students receive on average a 0.097 lower teacher-assigned grade than native students (column 1 of panel A), which corresponds to 0.09 standard deviations. In column 2, the gap between natives and immigrants is about one-half when teachers do not have implicit bias against immigrants ($IAT = 0$) compared to highly biased teachers ($IAT = 1$). However, the difference is not statistically significant, likely due to two factors: measurement error and bunching at the low end of the grade distribution. We next discuss these two issues in order.

First, for each teacher t, we calculate a standard measure of bias in grading (θ_t) obtained as the gap between native (n) and immigrant (i) students (n) in the difference between "nonblind" teacher-assigned grades (NB) and "blind" standardized test scores (B):

$$\theta_t = (NB_{nt} - B_{nt}) - (NB_{it} - B_{it}). \tag{1}$$

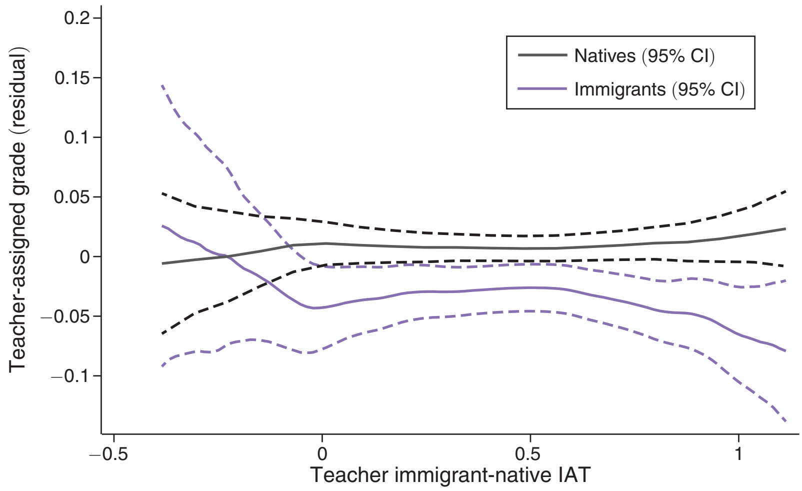

FIGURE 5. THE CORRELATION BETWEEN TEACHER-ASSIGNED GRADES AND THE IAT FOR IMMIGRANTS AND NATIVES

Notes: This graph shows the correlation between the residual of the teacher-assigned grade and the IAT for all teachers in the sample (math and literature). The residual is calculated absorbing the teacher fixed effects, a cubic polynomial of the INVALSI test score, and cohort fixed effects. The x-axis shows the immigrant-native IAT (raw *d*-score). The dotted lines represent the 95 percent confidence interval. Students in this sample completed grade 8 between school years 2011–2012 and 2015–2016.

In online Appendix Figure A.2, we correlate teachers' IAT with the above measure of bias in grading, calculated in two alternative ways. In the leftmost panel, we use a "naïve" measure that does not adjust for sample variation. This measure is positively but not significantly correlated with teachers' implicit bias (consistent with columns 2 and 3 of panel A of Table 3). In the rightmost panel of online Appendix Figure A.2, to reduce estimation error arising from sample variation, we calculate an empirical Bayes estimate of the bias in grading following Kane and Staiger (2002); Chetty, Friedman, and Rockoff (2014); and Terrier (2020).[22] This estimate shows a stronger positive and significant correlation with teachers' IAT score, suggesting that the measurement error, stemming from the fact that we only have data on a limited number of students for each teacher, may have large impacts on the results.

Second, the empirical relationship between bias in grading and the IAT score is mitigated by the bunching in end-of-semester grading at the pass grade (score 6), with more than 60 percent of immigrant students getting the pass grade in teacher-assigned evaluations (see Figure 3). This bunching makes it difficult to detect potential bias at the low end of the grade distribution. To gain more insights, we plot teacher-assigned grades by quintiles of the INVALSI score in Figure 6, separating teachers into high- and low-IAT groups (using 0.6 as the threshold for high bias, as in the literature). The figure shows that while teachers with low and high IAT scores give similar grades to native students throughout the test score distribution (right panel), teachers with stronger stereotypes give lower scores to high-performing immigrant students (left panel).

[22] Details on how we calculate the empirical Bayes estimate of the bias in grading are reported in online Appendix C.

Table 3—Bias in Grading and Teachers' IAT Scores

Teacher-assigned grade:	(1)	(2)	(3)
Panel A. All			
Immigrant	−0.097	−0.062	0.538
	(0.012)	(0.027)	(0.479)
IAT × Immigrant		−0.075	−0.065
		(0.050)	(0.049)
Observations	42,302	42,302	42,302
R^2	0.481	0.481	0.509
Panel B. High Ability			
Immigrant	−0.179	−0.115	1.635
	(0.020)	(0.042)	(0.772)
IAT × Immigrant		−0.139	−0.141
		(0.080)	(0.079)
Observations	25,415	25,415	25,415
R^2	0.403	0.403	0.442
Panel C. Low ability			
Immigrant	−0.056	−0.041	0.598
	(0.013)	(0.029)	(0.591)
IAT × Immigrant		−0.031	−0.031
		(0.060)	(0.058)
Observations	16,867	16,867	16,867
R^2	0.222	0.222	0.258
Teacher fixed effects	Yes	Yes	Yes
INVALSI cubic	Yes	Yes	Yes
Student controls	No	No	Yes
Student controls × Imm	No	No	Yes
Teacher controls × Imm	No	No	Yes

Notes: This table reports OLS estimates, where the dependent variable is the teacher-assigned grade. The unit of observation is student i taught by teacher t in school s. "Immigrant" indicates whether the student is not an Italian citizen. "IAT" indicates the Immigrant-Native IAT (d-score). Student controls include gender, generation of immigration, mother education, and province. Panel A provides the estimates for the full sample, panel B for high-ability students, and panel C for low-ability students, with a sample split based on the standardized test score INVALSI. Teacher controls include gender, place of birth, age, and age squared. Students in this sample completed grade 8 between school years 2011–2012 and 2015–2016. Standard errors are robust and clustered at the teacher level.

Column 1 of panels B and C in Table 3 show that the average gap in grading is three times as large for high-ability as for low-ability students. Furthermore, high-ability immigrant students get relatively lower grades than comparable native students when they are assigned to teachers with higher implicit stereotypes (columns 2 and 3 of panel B), while the gap is small and insignificant for low-ability immigrant students (columns 2 and 3 of panel C). Online Appendix Table A.8 shows that the results are qualitatively and quantitatively very similar when using the first difference between the teacher-assigned grades and test scores as an outcome.[23]

[23] In the field experiment, we collected two IAT scores with male and female names of natives and immigrants. Online Appendix Table D.1 reports the results using the gender-specific IAT for each students. The results are unaffected, as gender is not a focal characteristic of those IAT tests: the key categories are Immigrant and Native, and the brain is focused on those when completing the IAT (Greenwald and Banaji 1995).

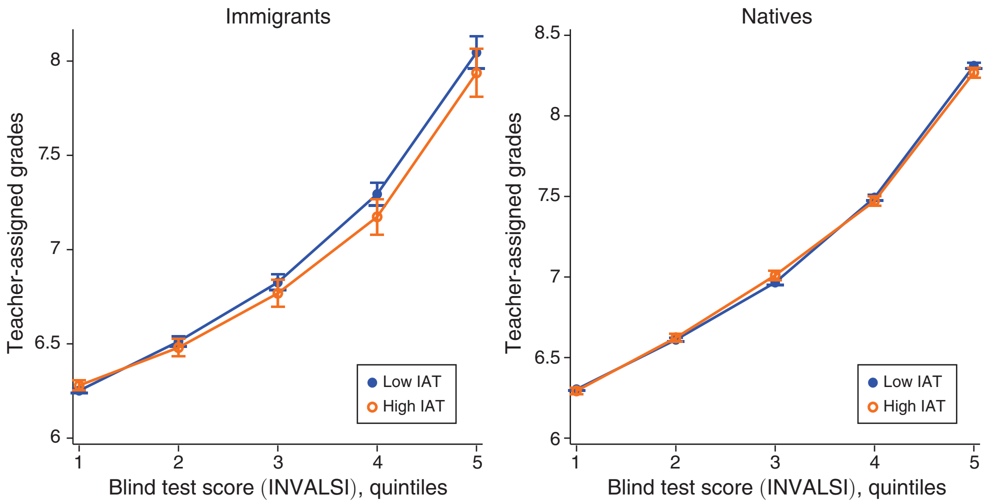

Figure 6. Teacher-Assigned Grades versus Blindly Graded Standardized Test Scores by Teacher IAT (High versus Low)

Notes: This graph shows teacher-assigned grades (nonblindly graded) on the vertical axis and quintiles of the standardized test score INVALSI (blindly graded) on the horizontal axis at the end of grade 8. Teacher-assigned grades are on a scale of 3 to 10, with 6 as the pass grade. The blue circles and lines are for students of teachers with an IAT lower than 0.6 (low bias), while the yellow circles and lines are for students of teachers with an IAT higher than 0.6 (high bias). The left panel presents grades for immigrant students, while the right panel presents grades for native students. Students in this sample completed grade 8 between school years 2011–2012 and 2015–2016.

IV. Main Results

A. *Field Experiment*

In the first experiment, we evaluate the effect that revealing to teachers their own stereotypes has on their grading behavior at the end of the first semester. Online Appendix Figure A.5 compares the distribution of grades assigned to immigrant and native students (left and right panel, respectively) by teachers in the treated (colored bar) and control groups (white bar). As explained in Section IIB, teachers randomized into the treated group were offered feedback on their IAT score before end-of-semester grading, while teachers randomized into the control group could receive the same information only after grading. The leftmost graph in online Appendix Figure A.5 shows that the distribution of grades assigned to immigrant students by teachers in the treated group shifts to the right compared to the distribution of those in the control group. The rightmost graph shows an opposite effect on grades assigned to native students.

In Table 4 we quantify the above effects by regressing the grade assigned by a teacher to a student on a treatment indicator for the teacher, a dummy for whether the student is immigrant, and the interaction between the two. Standard errors are clustered at the school level (the unit of randomization). Panel A shows the intention-to-treat effect: teachers offered the early IAT feedback exhibit a 0.35 point

Table 4—Impact of Revealing Stereotypes to Teachers in the Field Experiment

	Grade			Fail (Grade $<$ 6)		
Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Intention to treat						
IAT Feedback × Immigrant	0.351	0.369	0.367	−0.052	−0.059	−0.062
	(0.111)	(0.095)	(0.096)	(0.028)	(0.025)	(0.024)
Immigrant	−0.704	−0.683	0.294	0.118	0.088	−0.223
	(0.064)	(0.154)	(0.940)	(0.018)	(0.039)	(0.266)
IAT Feedback	−0.148	−0.166	−0.153	0.011	0.012	0.009
	(0.086)	(0.077)	(0.079)	(0.019)	(0.017)	(0.017)
Observations	10,279	10,279	10,279	10,279	10,279	10,279
R^2	0.028	0.126	0.131	0.012	0.043	0.047
Panel B. Local average treatment effect						
Email × Immigrant	0.450	0.471	0.466	−0.066	−0.074	−0.076
	(0.138)	(0.124)	(0.124)	(0.034)	(0.030)	(0.029)
Immigrant	−0.704	−0.632	0.245	0.118	0.080	−0.214
	(0.063)	(0.163)	(0.934)	(0.018)	(0.040)	(0.264)
Email	−0.200	−0.221	−0.202	0.015	0.016	0.012
	(0.114)	(0.106)	(0.107)	(0.026)	(0.022)	(0.022)
Observations	10,279	10,279	10,279	10,279	10,279	10,279
R^2	0.028	0.126	0.131	0.012	0.043	0.047
Mean control natives	7.03	7.03	7.03	0.10	0.10	0.10
Mean control immigrants	6.37	6.37	6.37	0.22	0.22	0.22
Students controls	No	Yes	Yes	No	Yes	Yes
Students controls × Imm	No	Yes	Yes	No	Yes	Yes
Teacher controls	No	No	Yes	No	No	Yes
Teacher controls × Imm	No	No	Yes	No	No	Yes

Notes: This table reports OLS estimates (panel A) and IV estimates (panel B), where the dependent variable is the grade (columns 1–3) or the probability of obtaining a grade lower than 6 (columns 4–6) at the end of the first semester of grade 8 (January). The unit of observation is student i in class c taught by teacher t in grade 8 of school s. Standard errors are robust and clustered at the school level. "IAT Feedback" is a dummy variable indicating whether the teacher was eligible for receiving the feedback before end-of-semester grading (January) or after end-of-semester grading (February). "Email" is a dummy variable indicating whether teachers eligible for receiving the feedback before end-of-semester grading actually requested it. The coefficients in panel B are estimated by instrumental variables, using "IAT Feedback" as an instrument for "Email." Student controls include gender, generation of immigration, and education of the mother, all interacted with whether the student is an immigrant. Teacher controls include gender, place of birth, age, and age squared, interacted with whether the student is an immigrant.

lower gap in grades between native and immigrant students (or 0.27 standard deviations) compared to teachers in the control group (column 1). The effect is generated by 0.2 point higher grades to immigrant students and 0.15 point lower grades to native students.

We interpret this result as driven by the implicit standardization of grades within each class and by the nature of the information provided to teachers. In fact, the IAT feedback compares the association with positive/negative attributes of native versus immigrant students. By virtue of randomization, the results are robust to controlling for student and teacher characteristics and the interaction of these characteristics with the *Immigrant* dummy (columns 2 and 3). As shown in online Appendix Figure A.6, the results are also robust to a permutation test that replicates specification (1) in Table 4 after randomly assigning the treatment variable *IAT Feedback* across teachers 1,000 times. In only 6 out of 1,000 cases, we find a coefficient larger than the one observed in Table 4.

A visual inspection of online Appendix Figure A.5 suggests that the effect may be particularly large around the margin that separates passing and failing students (i.e., between scores 6 and 5). This is confirmed in columns 4–6 of Table 4, where the dependent variable is the probability of failing. Early IAT feedback decreases the probability of failing immigrant students by about 6 percentage points, whereas failing rates of native students remain unaffected (the coefficient on the standalone *IAT Feedback* dummy is not significantly different from 0).

In panel B of Table 4, we rescale the intention-to-treat effect by the take-up rate of early IAT feedback, which was above 80 percent, to compute the treatment effect of stereotypes revelation. The variable *Email* in panel B of Table 4 equals one if the teacher *actually received* the feedback and zero if they did not receive any feedback. The coefficient on the interaction between the treatment and immigrant status increases in magnitude to about $+0.45$ for teacher-assigned grades (columns 1–3) and -0.07 for the probability of failing a school year.

Note that the magnitude of the treatment effect in Table 4 is not comparable to the magnitude of the bias in grading (i.e., the difference between teacher grades and standardized test scores) shown in Table 3. The reason is that the experiment was done at the end of the first semester (when standardized tests are not administered), while the bias in grading was measured at the end of the second semester (when we had information on both standardized test scores and teacher-assigned grades, and hence we can control for the standardized test score). Also, the grading policy of teachers typically differs between the first and the second semester, especially around the pass grade. Failing in the first semester represents a "warning" with no immediate consequences, while students may be retained in the same grade if they fail more than one subject in the second semester. For this reason, teachers are usually more reluctant to fail students in the second than in the first semester: indeed, the average fraction of students failing either literature or math (or both) is 21 percent in the first semester and only 2 percent in the second semester.[24]

Moreover, a lower propensity to fail students in the second semester sets a floor to teacher-assigned grades. For this reason, teachers' stereotypes likely induce a larger grade penalty for immigrant students in the first than in the second semester. To better compare the magnitude of the effects, we calculated a transition matrix between end-of-first-semester grades and end-of-second-semester grades for natives and immigrants.[25] We then estimated the impact of our field experiment using the transformed grade as the outcome. Online Appendix Table A.9 shows that, compared to our main result in panel A of Table 4, the magnitude of the intention-to-treat effect on the immigrant-native gap is reduced by 36 percent: the effect of revealing stereotypes to teachers is around 0.23 grade points, or 0.18 standard deviations.

[24] Among immigrant students, failure rates in the first and second semester reach 31 percent and 4 percent, respectively.

[25] Using the control schools, we calculate the transition matrix between end-of-first-semester grades and end-of-second-semester grades, separately for immigrants and natives. Then, for each student, we calculate the "transformed" grade as the average score of their in-group at the end of the second semester, conditional on end-of-first-semester grade.

Heterogeneous Effects.—We next investigate the heterogeneity in teacher response by the strength of the signal received, as measured by a variable equal to the average of the two IAT scores obtained by teachers. As explained in Section IIA, each teacher in our field experiment received two pieces of feedback: one for the IAT using male names of natives and immigrants and one for the IAT using female names. The information provided in our field experiment is therefore less precise compared to the online experiment, in which teachers received only one signal on their implicit stereotypes.

Figure 7 plots the local polynomial smooth of the native-immigrant grades on the IAT score of teachers.[26] Panel A refers to the field experiment, while panel B refers to the online experiment. The rightmost graph in panel A shows that in the control group, which received no additional information before grading, native students receive higher grades compared to immigrants. The leftmost graph shows that the difference in the residuals is close to zero for the teachers who received the IAT feedback. Although imprecisely estimated, the slope in the relationship between the residualized grade gap and the IAT score is positive for control teachers and negative for treated ones. This qualitatively suggests that the most biased teachers are more generous toward natives compared to immigrants, but they change their behavior more when receiving the information on their own IAT score.

In Table 5 we analyze the same heterogeneity in regression format. Columns 1–2 refer to the field experiment, and columns 3–4 to the online one. Column 1 reports the baseline estimated effects, using the most stringent specification in column 3 of Table 4. Column 2 quantifies the evidence in panel A of Figure 7. The coefficient on the triple interaction (*IAT Feedback* × *Immigrant* × *IAT Score*) in column 2 of Table 5 is positive, suggesting that the treatment induced more generous grading toward immigrants for teachers with stronger negative stereotypes, but the result is imprecisely estimated.

A higher average IAT encompasses two features. First, other things equal, teachers with a higher IAT receive a stronger signal about their implicit biases, which should in principle induce a greater reaction. Second, these teachers may be less willing to adjust their behavior, precisely because they are more biased to start with. The results we discussed capture both these effects. The online experiment will allow us to delve deeper into the differential effect of signal strength using a measure of the unexpected component of bias revelation. In addition, participants in the online experiment conducted only one IAT and received thus only one feedback, which may lead to a more precise reaction to their own IAT score. Teachers in the field experiment, by contrast, conducted two IAT tests (one with male and one with female names), and it is ex ante unclear whether they would respond to the average of the two scores or to only one of them.[27]

[26] More precisely, the figure plots the difference in the mean residuals of natives and the mean residuals of immigrants for the teachers. First, we calculate the residuals of the grade for each student absorbing our simple set of baseline controls (gender, education, and occupation of parents). Then, for each teacher we calculate the difference in the weighted mean of residuals of natives and the weighted mean of residuals of immigrants.

[27] We also tried using the gender-specific IAT score in the regression, depending on the gender of the student, but we found that the results presented in column 2 of Table 5 were unaffected (see column 2 of online Appendix Table D.2). This is not necessarily surprising because by construction, the IAT score should depend only on the categories (Immigrant-Native) and not on other nonfocal characteristics (e.g., the gender in our case).

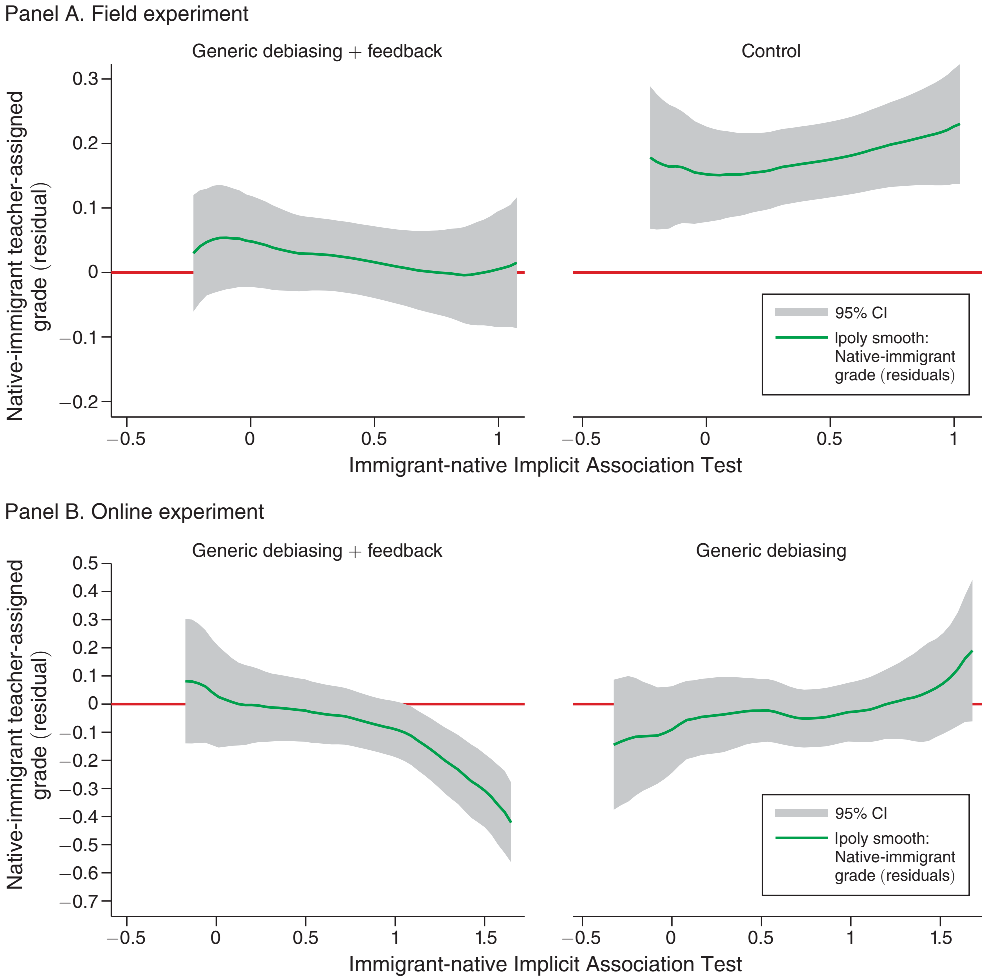

FIGURE 7. THE IMPACT OF REVEALING STEREOTYPES TO TEACHERS ON GRADING

Notes: This graph shows the difference in grading of teachers in the field experiment (top panel) and online experiment (bottom panel) by their immigrant-native IAT score (raw *d*-score). First, we calculate for each grade given by teachers the residual considering the standard set of controls (for the field experiment: gender, education and occupation of parents, teacher controls such as gender, age, place of birth; for the online experiment: original grade on the question, subject, and order of questions). Then, for each teacher in our sample, we calculate the difference between the weighted mean of residuals of natives and the mean of residuals of immigrants.

In online Appendix Table A.10, we explore the heterogeneity of the (intention-to-treat) effect along other teacher characteristics. The first is teachers' explicit bias against immigrants. Learning that one holds negative (implicit) stereotypes against immigrants conveys more information to teachers who are unaware of such stereotypes, and hence we should expect a stronger reaction from teachers who reported no explicit bias in the baseline survey. To test this hypothesis, in column 2 of online Appendix Table A.10, we include a triple interaction between the indicator for immigrant student, early IAT feedback, and the dummy variable "*WVS*," which

Table 5—The Impact of Revealing Stereotypes in the Field and Online Experiment, by Teacher IAT Score

	Field experiment		Online experiment	
Teacher-assigned grade:	(1)	(2)	(3)	(4)
Immigrant	0.294	0.371	0.109	0.420
	(0.940)	(0.925)	(0.083)	(0.141)
IAT Feedback	−0.153	−0.122	−0.216	0.005
	(0.079)	(0.115)	(0.111)	(0.173)
IAT Feedback × Immigrant	0.367	0.267	0.017	−0.580
	(0.096)	(0.174)	(0.122)	(0.196)
IAT Feedback × IAT Score		−0.065		−0.329
		(0.154)		(0.171)
IAT Feedback × Immigrant × IAT Score		0.214		0.849
		(0.302)		(0.241)
IAT Score		0.019		−0.068
		(0.122)		(0.130)
Immigrant × IAT Score		−0.078		−0.426
		(0.225)		(0.157)
Control mean	6.944	6.944	7.134	7.134
Observations	10,279	10,279	1,460	1,460
R^2	0.131	0.131	0.440	0.450
Subject, order, original grade FE	No	No	Yes	Yes
Student controls	Yes	Yes	Yes	Yes
Teacher controls	Yes	Yes	Yes	Yes

Notes: This table reports OLS estimates, where the dependent variable is the grade assigned by teachers in the field experiment in columns 1–2 and online experiment in columns 3–4. The unit of observation is student *i* by teacher *t*. Standard errors are robust and clustered at the school level (the unit of randomization). "IAT Feedback" is a dummy variable indicating whether the teacher was eligible for receiving the IAT feedback versus the active control message. "IAT Score" is a continuous variable indicating the standard *d*-score of the Immigrant-Native IAT test (more details available in online Appendix B). Student controls include gender, generation of immigration, and education of the mother, all interacted with whether the student is an immigrant for the field experiment. Teacher controls include gender, place of birth, age, and age squared, interacted with whether the student is an immigrant for the field experiment. Student controls include gender and class for the online experiment. Teacher controls include gender, place of birth, and a dummy for whether the teacher completed the IAT before the first reminder for the online experiment.

equals one for teachers who agree with the statement that "immigrants and natives should have equal opportunities to access available jobs." The positive and significant coefficient on the triple interaction confirms that this group is more responsive to the intervention, consistent with the fact that they may have been less aware of their (implicit) stereotypes before our treatment.

We next explore the role played by awareness of anti-immigrant bias in society. Carlana, La Ferrara, and Pinotti (2022b) show that teachers with stronger implicit bias are more likely to recommend vocational tracks and less likely to recommend top-tier tracks to immigrant students. In our survey we asked teachers whether they believed that bias against immigrant students may be why they enroll disproportionately into less demanding high school tracks compared to natives with the same performance. Twenty percent of teachers answered that it was "likely" or "extremely likely" that prejudice affected the choice of immigrant students. Column 3 in online Appendix Table A.10 shows that these same teachers react more strongly to receiving information on their own implicit bias.

B. *Online Experiment*

As explained in Section IIC, we conducted a second experiment in which a different group of teachers was asked to grade ten tests, with randomly assigned native- or immigrant-sounding student names to each test. As in the first experiment, teachers took an IAT at baseline, and we provided feedback on the IAT result only to a random group. Different from the first experiment, however, both the treated and the (active) control group received a generic debiasing message.

We start by visually showing the results of this experiment in panel B of Figure 7. The two graphs plot the average difference in grades assigned to native and immigrant students against teacher IAT scores, controlling for the quality of the answer, exam order, and subject.[28] The leftmost graph shows that receiving feedback on one's own IAT, in addition to the generic debiasing message, reduces the native-immigrant gap in grades for teachers who display relatively high levels of implicit bias, and the reduction is larger the higher their bias (IAT score). This is consistent with the interpretation that teachers who receive a more negative signal react by helping immigrants more. In contrast, there is a weakly positive—but insignificant—relationship between the gap in grades and IAT scores across teachers who receive only the debiasing message (rightmost graph).

Column 3 of Table 5 reports the main results of the online experiment in regression format. On average, receiving personalized feedback leads to a small decrease in the grades assigned across the board (significant at the 10 percent level), but it does not lead to a relative increase in grades assigned to immigrant students *compared to the debiasing message*. Recall that in the field experiment we observed a decrease in grading bias against immigrants as a consequence of the IAT feedback *compared to no information* in the control group. The two results thus provide interesting, complementary evidence: the two policies—generic debiasing message and personalized IAT feedback—have similar effects on average, but Figure 7 clearly shows heterogeneous effects depending on the feedback the teachers received.

Column 4 quantifies the results from panel B of Figure 7. Teachers with no stereotypes ($IAT\ Score = 0$), who receive a generic debiasing message but remain unaware of their own IAT ($Feedback = 0$), assign a grade 0.42 points higher to immigrant students than to native ones. The fact that raising teachers' general awareness may reduce biased behavior is consistent with previous evidence on debiasing interventions (Boring and Philippe 2021). The positive correlation on the grade of immigrant students disappears for teachers with an IAT equal to one (the coefficient on $Immigrant \times IAT\ Score$ is -0.426, with a standard error of 0.157). Teachers with IAT greater than one assign lower grades to immigrant students compared to native ones in the "active control" group.

What happens to teachers' grading when they receive information on their own implicit stereotypes on top of the general debiasing message? The effect varies along the distribution of the IAT score. Teachers essentially assign the same grades, on average, to immigrant students and native students when the feedback reveals the

[28] Recall from Section IIC that the answers to the tests that the teachers graded were prepared by consultants who also provided a score for each potential answer. This is the variable we include among the regressors to control for the "quality" of the answer.

absence of stereotypes (i.e., when *IAT Score* $= 0$, the gap in grading for immigrant students relative to native students is $0.420 - 0.580 = -0.16$ points, statistically indistinguishable from 0), but they significantly increase grades to immigrant students when the feedback reveals strong implicit stereotypes. The higher the IAT score—and therefore, the signal received about one's own implicit bias—the stronger the teachers' response.

When receiving feedback on their IAT score, teachers are informed on whether they have "no/slight/moderate/severe" stereotypes against immigrants. Column 1 of Table 6 replicates column 3 of Table 5 using a dummy variable taking value one if the teacher is "moderately or severely" biased, instead of the continuous IAT score.[29] The results are confirmed: teachers have a significantly stronger positive reaction in favor of immigrants when receiving the information that they have moderate/severe implicit stereotypes.

Finally, we create an indicator for teachers underestimating their own bias, which equals one if the feedback received by the teacher is more negative than their prior (elicited at baseline using the same ordinal scale).[30] Column 2 of Table 6 shows that the effect of revealing stereotypes is driven by teachers who *underestimate* their own IAT and were thus "surprised" by the information received. The coefficient of the triple interaction (*IAT Feedback* $\times$ *Immigrant* $\times$ *Underestimate Own IAT*) is unaffected when we include teachers' IAT score among the regressors (column 3) to account for the mechanical correlation due to more biased teachers being more likely to underestimate their own IAT.

In the last column of Table 6, we include all interactions from columns 1 and 2 to jointly investigate the role of the severity of the IAT feedback received and the belief on own IAT. The two variables are highly correlated (correlation coefficient of 0.71): 94 percent of the teachers with moderate or severe IAT also underestimate the feedback they receive. The coefficients on the two triple interactions in column 4 remain positive but are less precisely estimated.[31] To sum up, teachers with higher IAT and teachers who update their beliefs more as a result of the signal change their grading behavior to a larger extent, but the data do not allow us to perfectly disentangle the two driving factors, as the two variables are highly correlated.

V. Conclusions

Immigrant students receive lower teacher-assigned grades than native students after controlling for their performance on blindly graded standardized tests. The gap is substantially wider for high-achieving immigrant students. We acknowledge that there may be characteristics that differentiate immigrant students from native students that are observable to teachers but unobservable to the econometrician (e.g., disciplinary problems or differences in performance on standardized multiple-choice questions versus open-ended ones). We show that for high-ability

[29] In the online experiment, 80.8 percent of teachers have an IAT score above 0.35 and therefore are "moderately or severely" biased.

[30] In the online experiment, 81.3 percent of teachers underestimate their own bias.

[31] As shown in online Appendix Table A.11, the results are very similar when considering the continuous measures—instead of dummies—for IAT score and the difference between the own score and the expected score.

TABLE 6—BELIEFS UPDATING IN THE ONLINE EXPERIMENT

Teacher-assigned grade:	(1)	(2)	(3)	(4)
IAT Feedback	0.042	−0.056	−0.056	−0.009
	(0.205)	(0.281)	(0.274)	(0.264)
Immigrant	0.400	0.401	0.423	0.463
	(0.173)	(0.161)	(0.178)	(0.191)
IAT Feedback × Immigrant	−0.475	−0.514	−0.515	−0.591
	(0.230)	(0.237)	(0.242)	(0.249)
Moderate/Severe IAT	−0.064		−0.196	−0.087
	(0.161)		(0.172)	(0.211)
IAT Feedback × Moderate/Severe IAT	−0.317			−0.219
	(0.228)			(0.335)
Immigrant × Moderate/Severe IAT	−0.358		−0.097	−0.270
	(0.184)		(0.222)	(0.258)
IAT Feedback × Immigrant × Moderate/Severe IAT	0.608			0.345
	(0.262)			(0.429)
Underestimate own IAT		−0.069	0.074	−0.005
		(0.226)	(0.265)	(0.290)
IAT Feedback × Underestimate own IAT		−0.198	−0.199	−0.038
		(0.305)	(0.297)	(0.424)
Immigrant × Underestimate own IAT		−0.354	−0.285	−0.160
		(0.172)	(0.225)	(0.231)
IAT Feedback × Immigrant × Underestimate own IAT		0.660	0.661	0.409
		(0.271)	(0.275)	(0.439)
Control mean	7.134	7.134	7.134	7.134
Observations	1,460	1,460	1,460	1,460
R^2	0.446	0.448	0.450	0.450
Subject, order, original grade fixed effects	Yes	Yes	Yes	Yes
Student controls	Yes	Yes	Yes	Yes
Teacher controls	Yes	Yes	Yes	Yes

Notes: This table reports OLS estimates, where the dependent variable is the grade assigned by teachers in the online experiment. The unit of observation is student i by teacher t. Standard errors are robust and clustered at the school level (the unit of randomization). "IAT Feedback" is a dummy variable indicating whether the teacher was eligible for receiving the IAT feedback versus the active control message. "Moderate/Severe IAT" is a dummy variable indicating whether the IAT is above 0.35 and the teachers received as feedback a moderate or severe association Immigrant-Bad Native-Good. "Underestimate own IAT" is a dummy that equals one if the teacher believes her IAT score is lower compared to the actual score. Student controls include gender and class. Teacher controls include gender, place of birth, and a dummy for whether the teacher completed the IAT before the first reminder.

students, the difference in the grading of native and immigrant students is systematically correlated with teachers' stereotypes against immigrants, a pattern strongly suggestive of bias.

We conduct two novel experiments to test whether informing teachers about their own stereotypes may be an effective policy to reduce discrimination in grading. Our main treatment consists of receiving feedback on one's own IAT score. In the first experiment ("field experiment"), we share the IAT feedback with teachers in the treated group just before end-of-term grading, i.e., in time to adjust the grade given to students. The control group receives feedback right after end-of-term grading, i.e., too late to adjust grades. We find that teachers (randomly) assigned to the treatment react to the information by increasing the grades they give to immigrant students and decreasing the grades they give to native students. The effect is particularly strong around the threshold that determines whether a student passes or fails a subject.

In the second experiment ("online experiment"), teachers in the (active) control group receive a generic debiasing message, while teachers in the treatment group receive the debiasing message plus information on their own IAT score. Three weeks later, both groups are asked to grade tests randomly assigned to native- or immigrant-sounding names. We find that, on average, informing teachers of their own stereotypes does not increase grades given to immigrant students relative to natives, compared to the active control group. However, we find important heterogeneity based on teachers' baseline IAT. When teachers receive only generic debiasing, the higher their implicit stereotypes, the lower the grade assigned to immigrant students compared to native ones. When teachers receive information on their own stereotypes, they significantly increase the grades given to immigrant students compared to those given to natives only when their feedback suggests they hold negative views against immigrants. Furthermore, thanks to the elicitation of teachers' priors about their own IAT, we can show that the effect is driven by teachers who did not expect to receive negative feedback. This suggests that the effect of revealing stereotypes may come mainly from people being unaware of their own implicit bias.

Our findings can help inform an active policy debate regarding recent efforts by corporations, universities, schools, and other institutions to increase awareness about implicit bias by encouraging search committee members or new employees to take an IAT. In the context of schooling, the IAT is simple to implement, and it would not cost much to ask every teacher to take it, say, at the beginning of the academic year.[32] This may help counteract negative stereotypes about certain groups. However, the implications of such a policy are not straightforward. By making teachers aware of their "implicit" biases, their evaluation of students becomes fairer if they were acting upon their stereotypes by giving lower grades to immigrants. But it is possible that teachers whose negative stereotypes do not translate into discriminatory behavior may also react, thus inducing positive discrimination toward immigrant children. Further research on this point is warranted.

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